

BEC303 Network Analysis — Assignment 3 — Initial conditions, time constants and second-order response

Module 3: Transient Behavior and Initial Conditions

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO3: 13 · L1–L2 0, L3–L4 9, L5–L6 4

Objective

Make initial- and final-condition evaluation automatic, establish fluency with first-order time constants, and develop correct classification and solution of second-order circuits under both DC and AC excitation.

Learning outcomes

- Student will be able to evaluate all initial and final conditions, including derivatives, without writing a differential equation.
- Student will be able to compute the correct time constant from the post-switching resistance.
- Student will be able to classify and solve overdamped, critically damped and underdamped responses.
- Student will be able to analyse sequentially switched circuits and AC switching transients.

Part A — Problems (17 marks)

P1. For a circuit of your construction containing one capacitor and one inductor and at least one switch, tabulate every branch voltage and current at $t = 0^-$, $t = 0^+$ and $t = \infty$. Mark clearly which two quantities were carried across by continuity. [C03 | L3 | 2 marks]

P2. For the same circuit, compute dv_C/dt and di_L/dt at $t = 0^+$ from the element laws, and state why these cannot be obtained from the $t = 0^-$ solution. [C03 | L4 | 2 marks]

P3. Find the time constant of an RL circuit in which the resistance seen by the inductor differs before and after switching. Show both values and explain which one governs the decay. [C03 | L3 | 2 marks]

P4. Plot a measured or simulated exponential decay on both linear and logarithmic vertical axes, extract τ by the tangent method and by the semilog slope, and compare the two results. [C03 | L4 | 2 marks]

P5. Apply a rectangular pulse of width 2τ to an initially uncharged RC circuit. Obtain the full response by superposing two step responses and verify the handover value at the end of the pulse. [C03 | L4 | 2 marks]

P6. Design an RC power-ON delay for a stated supply, threshold and delay time. Choose standard component values, recompute the actual delay, and tabulate the delay spread when supply and threshold

each vary by $\pm 10\%$. [C03 | L5 | 2 marks]

P7. Analyse a circuit with three switching instants where at least one interval is shorter than 3τ . Present the response as an explicit piecewise function and state each handover value. [C03 | L4 | 2 marks]

P8. For a parallel RLC of your choice, compute α and ω_0 , classify the damping, then change R so that the circuit becomes critically damped. State the required R and verify the classification. [C03 | L4 | 1 marks]

P9. Solve a step-driven series RLC completely, obtaining both natural-response constants from the value and the derivative at $t = 0+$, and verify that your expression satisfies both conditions. [C03 | L4 | 1 marks]

P10. Compute the switching transient for an RL load connected to an AC supply at both a voltage zero and a voltage peak, and report the ratio of the two first-peak currents. [C03 | L5 | 1 marks]

Part B — Theory (3 marks)

T1. Explain why only capacitor voltage and inductor current are continuous across a switching instant, and describe one physical consequence of each rule that an engineer must design around. [C03 | L4 | 1 marks]

T2. Argue why the 5τ settling convention is inadequate for a 12-bit data acquisition system, and state the number of time constants actually required. [C03 | L5 | 1 marks]

T3. Explain why increasing R damps a series RLC but undamps a parallel RLC, using the position of the resistance in the energy path rather than the formulas alone. [C03 | L5 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC303_Assignment3.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Correct initial and final conditions	6	All three time points tabulated correctly with continuity applied only to the two valid quantities and derivatives computed from element laws.	Values correct with minor gaps in the derivative evaluation.	Initial and final values mostly correct but continuity applied loosely.	Non-state quantities carried across the switching instant; $t = 0+$ and $t = \infty$ equivalents confused.
Time constants and response forms	6	Every τ computed from the correct post-switching resistance; every response form matches its damping classification.	Correct forms with one minor τ error.	Some τ values taken from the pre-switching topology.	Wrong solution form used, e.g. no t factor in the critically damped case.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Sequential and AC switching analysis	4	Handover values computed from actual interval end values; AC transient correctly decomposed into forced and natural parts with the switching angle addressed.	Correct chaining with a minor slip in one interval.	Intervals chained but asymptotic values used for handover.	Each interval restarted from zero or from the supply value.
Verification, plots and engineering commentary	4	Every result checked against a limiting case or an independent method; plots correctly labelled with units; design tolerances quantified.	Most results checked; plots labelled.	Occasional checks; plots present but sparsely labelled.	No verification; unlabelled plots; tolerance ignored.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO3	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.